

AIIMS CRE EXAM PREPARATION

Anesthesia & OT Technician (Post Code 11) - Memory Based Questions

Q1. What does a square wave pattern on an EtCO₂ capnograph indicate during general anesthesia?

- A) Spontaneous breathing
- B) Mechanical ventilation
- C) Cardiac arrest
- D) Bronchospasm

Correct Answer: B) Mechanical ventilation

Q2. Which of the following airway devices is used in difficult intubation to lift the epiglottis with a movable tip?

- A) Macintosh laryngoscope
- B) McCoy laryngoscope
- C) Miller laryngoscope
- D) ProSeal LMA

Correct Answer: B) McCoy laryngoscope

Q3. Which needle is used to create pneumoperitoneum via gas insufflation in laparoscopic surgeries?

A) Tuohy needle

B) Spinal needle

C) Veress needle

D) Huber needle

Correct Answer: C) Veress needle

Q4. What is the safety mechanism used to prevent incorrect cylinder attachment on an anesthesia machine?

A) Diameter Index Safety System (DISS)

B) Pin Index Safety System (PISS)

C) Bodok seal

D) APL valve

Correct Answer: B) Pin Index Safety System (PISS)

Q5. What is the oxygen percentage inside a pure oxygen cylinder if the pressure gauge shows 20 bar?

- A) 21%
- B) 50%
- C) 75%
- D) 100%

Correct Answer: D) 100%

Q6. What is the color coding of an oxygen cylinder?

- A) Blue body, white shoulder
- B) Black body, white shoulder
- C) Black body, grey shoulder
- D) French blue

Correct Answer: B) Black body, white shoulder

Q7. What is the standard color of a Nitrous Oxide (N₂O) cylinder?

- A) Black
- B) White
- C) Blue
- D) Grey

Correct Answer: C) Blue

Q8. What is the normally used Endotracheal (ET) tube size for an adult male?

A) 6.0 - 6.5 mm

B) 7.0 - 7.5 mm

C) 8.0 - 8.5 mm

D) 9.0 - 9.5 mm

Correct Answer: C) 8.0 - 8.5 mm

Q9. Which of the following is the shortest-acting local anesthetic?

A) Bupivacaine

B) Lignocaine

C) Ropivacaine

D) Chlorprocaine

Correct Answer: D) Chlorprocaine

Q10. Which is the shortest-acting opioid analgesic commonly used in anesthesia?

A) Fentanyl

B) Morphine

C) Remifentanyl

D) Sufentanyl

Correct Answer: C) Remifentanyl

Q11. What is the standard port size for a laparoscopic camera?

A) 2 mm

B) 5 mm

C) 10 mm

D) 15 mm

Correct Answer: C) 10 mm

Q12. Which vein is considered the vein of choice for cannulation in pediatric patients?

A) Internal jugular vein

B) Subclavian vein

C) Dorsum of the hand

D) Femoral vein

Correct Answer: C) Dorsum of the hand

Q13. Why is Nitrous Oxide (N₂O) generally avoided in laparoscopic procedures and closed cavity surgeries?

- A) It is highly flammable
- B) It expands in air-filled cavities due to high solubility
- C) It causes severe bradycardia
- D) It decreases intra-abdominal pressure

Correct Answer: B) It expands in air-filled cavities due to high solubility

Q14. An increase in Positive End-Expiratory Pressure (PEEP) directly affects and increases which lung volume?

- A) Tidal Volume (TV)
- B) Inspiratory Reserve Volume (IRV)
- C) Functional Residual Capacity (FRC)
- D) Vital Capacity (VC)

Correct Answer: C) Functional Residual Capacity (FRC)

Q15. In obstructive lung diseases, how is the FEV1/FVC ratio affected?

- A) Increases
- B) Decreases
- C) Remains normal
- D) Becomes 100%

Correct Answer: B) Decreases

Q16. What is the normal anatomical dead space in an adult human?

- A) 50 ml
- B) 100 ml
- C) 150 ml
- D) 250 ml

Correct Answer: C) 150 ml

Q17. 1 gram of fully saturated hemoglobin can carry approximately how much oxygen?

- A) 1.00 ml
- B) 1.34 ml
- C) 2.50 ml
- D) 5.00 ml

Correct Answer: B) 1.34 ml

Q18. Which is the induction agent of choice for hemodynamically unstable patients?

A) Propofol

B) Thiopentone

C) Etomidate

D) Ketamine

Correct Answer: C) Etomidate

Q19. Which biological indicator is used to verify the effectiveness of steam sterilization (Autoclaving)?

A) *Bacillus subtilis*

B) *Geobacillus stearothermophilus*

C) *Bacillus atrophaeus*

D) *Clostridium tetani*

Correct Answer: B) *Geobacillus stearothermophilus*

Q20. What is the primary function of the APL (Adjustable Pressure Limiting) valve in an anesthesia machine?

- A) To mix oxygen and nitrous oxide
- B) To vaporize the anesthetic agent
- C) To adjust and limit the pressure of gas in the breathing circuit
- D) To monitor ETCO₂

Correct Answer: C) To adjust and limit the pressure of gas in the breathing circuit

Q21. Which drug produces 'dissociative anesthesia'?

- A) Propofol
- B) Dexmedetomidine
- C) Ketamine
- D) Midazolam

Correct Answer: C) Ketamine

Q22. What is the ideal recommended temperature for an Operation Theater (OT)?

A) 15 - 18°C

B) 20 - 24°C

C) 25 - 28°C

D) 30 - 32°C

Correct Answer: B) 20 - 24°C

Q23. What is the oxygen flush volume delivered by the anesthesia machine?

A) 10-15 L/min

B) 20-30 L/min

C) 35-75 L/min

D) 100-120 L/min

Correct Answer: C) 35-75 L/min

Q24. Why is a cautery plate (grounding pad) applied to the patient during electrosurgery?

- A) To monitor heart rate
- B) To neutralize the current and prevent electrical burn injuries
- C) To keep the patient warm
- D) To increase anesthetic absorption

Correct Answer: B) To neutralize the current and prevent electrical burn injuries

Q25. Which of the following is considered the 'gold standard' for confirming correct endotracheal intubation?

- A) Chest rise
- B) Auscultation of lungs
- C) Pulse oximetry
- D) Capnography (EtCO₂)

Correct Answer: D) Capnography (EtCO₂)

Q26. What is the narrowest portion of the airway in a child less than 6 years old?

- A) Glottis
- B) Subglottis (Cricoid ring)
- C) Trachea
- D) Carina

Correct Answer: B) Subglottis (Cricoid ring)

Q27. Which circuit is most commonly preferred for pediatric ventilation?

- A) Mapleson A (Magill)
- B) Mapleson D (Bain)
- C) Mapleson F (Jackson Rees)
- D) Circle system

Correct Answer: C) Mapleson F (Jackson Rees)

Q28. Local anesthetics primarily function by blocking which ion channels?

- A) Calcium channels
- B) Potassium channels
- C) Sodium channels
- D) Chloride channels

Correct Answer: C) Sodium channels

Q29. Minimum Alveolar Concentration (MAC) is NOT increased in which of the following conditions?

A) Hyperthermia

B) Hypernatremia

C) Chronic alcoholism

D) Old age

Correct Answer: D) Old age

Q30. Which device is used in the breathing circuit to moisturize and warm the anesthetic gases?

A) Vaporizer

B) Flowmeter

C) HME (Heat and Moisture Exchanger) filter

D) Scavenger

Correct Answer: C) HME (Heat and Moisture Exchanger) filter

Q31. Which of the following is an example of an absorbable suture?

- A) Silk
- B) Prolene
- C) Nylon
- D) Vicryl (Polyglactin 910)

Correct Answer: D) Vicryl (Polyglactin 910)

Q32. Which of the following IV fluids is a crystalloid?

- A) Albumin
- B) Dextran
- C) Normal Saline (NS)
- D) Gelatin

Correct Answer: C) Normal Saline (NS)

Q33. Cormack and Lehane grading system is used for what purpose?

- A) Grading surgical wounds
- B) Visualizing the glottis during direct laryngoscopy
- C) Classifying physical status of patients (ASA)
- D) Assessing depth of anesthesia

Correct Answer: B) Visualizing the glottis during direct laryngoscopy

Q34. What is the recommended energy dose for the first shock in the treatment of pulseless ventricular tachycardia (VT)?

A) 50-100 Joules

B) 120-200 Joules

C) 300 Joules

D) 360 Joules

Correct Answer: B) 120-200 Joules

Q35. A 'curare cleft' seen on a capnography waveform indicates:

A) Malignant hyperthermia

B) Esophageal intubation

C) Inadequate muscle relaxation

D) Complete airway obstruction

Correct Answer: C) Inadequate muscle relaxation

Q36. What is the correct anatomical order of a surgical hand scrub?

A) Scrub -> Wet -> Dry -> Rinse

B) Wet -> Scrub -> Rinse -> Dry

C) Rinse -> Wet -> Scrub -> Dry

D) Scrub -> Rinse -> Wet -> Dry

Correct Answer: B) Wet -> Scrub -> Rinse -> Dry

Q37. Which of the following procedures does NOT increase anatomical dead space?

- A) Using a face mask
- B) Adding a catheter mount
- C) Bronchodilation
- D) Endotracheal intubation

Correct Answer: D) Endotracheal intubation

Q38. During general anesthesia with a muscle relaxant, the vocal cords assume which position?

- A) Fully open (abducted)
- B) Fully closed (adducted)
- C) Cadaveric position (midline)
- D) Paramedian position

Correct Answer: C) Cadaveric position (midline)

Q39. Which law states that pressure and volume of a gas are inversely proportional at a constant temperature?

A) Charles's Law

B) Boyle's Law

C) Dalton's Law

D) Graham's Law

Correct Answer: B) Boyle's Law

Q40. Where does the fresh gas flow go immediately after passing the flowmeters in the anesthesia machine?

A) Breathing circuit

B) Oxygen flush valve

C) Vaporizer

D) Scavenging system

Correct Answer: C) Vaporizer

Q41. What does a completely flat capnography line immediately after intubation suggest?

- A) Successful intubation
- B) Esophageal intubation, apnea, or cardiac arrest
- C) Hyperventilation
- D) Hypoventilation

Correct Answer: B) Esophageal intubation, apnea, or cardiac arrest

Q42. What is the recommended compression-to-ventilation ratio for CPR in a pediatric patient with TWO rescuers?

- A) 30:2
- B) 15:2
- C) 30:1
- D) 15:1

Correct Answer: B) 15:2

Q43. What is the appropriate immediate management for foreign body airway obstruction in a conscious infant?

- A) Abdominal thrusts
- B) Blind finger sweep
- C) 5 Back blows and 5 chest thrusts
- D) Tracheostomy

Correct Answer: C) 5 Back blows and 5 chest thrusts

Q44. Which of the following IV agents is primarily used in TIVA (Total Intravenous Anesthesia)?

- A) Sevoflurane
- B) Isoflurane
- C) Propofol
- D) Halothane

Correct Answer: C) Propofol

Q45. Which of the following is considered an osmotic diuretic/laxative?

A) Furosemide

B) Mannitol

C) Spironolactone

D) Hydrochlorothiazide

Correct Answer: B) Mannitol

Compiled from memory-based exam analysis for AIIMS CRE.